

AMENDMENT NO. 1 SEPTEMBER 2024
TO
IS 5484 : 2023 EC GRADE ALUMINIUM ROD PRODUCED BY CONTINUOUS
CASTING AND ROLLING — SPECIFICATION
(Third Revision)

(Page 2, clause 10, Title) — Substitute the following for the existing:

‘ELECTRICAL RESISTIVITY AND ELECTRICAL CONDUCTIVITY’.

(Page 2, clause 10.1) — Substitute the following for the existing:

‘10.1 Electrical resistivity at 20 °C shall be determined on representative sample by resistance measurement as per method mentioned in Annex C. Electrical conductivity shall be calculated by the formula mentioned below or measured by eddy current method mentioned in IS 5082:

$$\text{Electrical conductivity percent, IACS} = \frac{0.017\,241 \times 100}{(\text{Electrical resistivity at } 20\,^{\circ}\text{C})}$$

where unit of electrical resistivity is $\Omega\cdot\text{mm}^2/\text{m}$.’

(Page 2, clause 10.1.1) — Substitute the following for the existing:

‘10.1.1 At a temperature of 20 °C the electrical resistivity and electrical conductivity of rods as per corresponding temper designation shall be as given in Table 2.’

(Page 5, Annex C) — Insert Annex C after Annex B:

‘ANNEX C

(Clause 10.1)

MEASUREMENT OF ELECTRICAL RESISTIVITY

C-1 SCOPE

This test method covers the determination of the electrical resistivity of aluminium and aluminium alloys in various forms (wire, strip, rod, bar).

C-2 DEFINITIONS

- a) **Resistivity** — Resistivity is the electrical resistance of a body of unit length, and unit cross-sectional area or unit weight; and
- b) **Volume Resistivity** — Volume resistivity is commonly expressed in ohms for a theoretical conduct or of unit length and cross-sectional area in acceptable metric units in $\Omega\cdot\text{mm}^2/\text{m}$. It may be calculated by the following equation:

$$p_v = \left(\frac{A}{L}\right) R$$

where

p_v = volume resistivity, in $\Omega\cdot\text{mm}^2/\text{m}$;

A = cross-sectional area, in mm^2 ;

L = gauge length, in mm; and

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R = measured resistance, in Ω .

C-3 APPARATUS

Resistance shall be measured with a circuit configuration and instrumentation that has a resistance measurement capability of ± 0.15 percent accuracy.

C-4 TEST SPECIMEN

- The test specimen may be in the form of a wire, strip, rod, bar, tube, or shape. It shall be of uniform cross section throughout its length within ± 0.75 percent of the cross-sectional area. Wherever possible it shall be the full cross section of the material it represents, if the full cross section is such that the uniformity of the cross-sectional area can be accurately determined. A test length of at least 1 ft or 300 mm;
- Cross-sectional dimensions of the specimen may be determined by micrometer measurements with required accuracy of ± 0.10 percent, if required accuracy not possible with micrometer, determine the cross-section from weight, density, and length of the specimen (for purpose of calculation take density of aluminium: 2.705 g/cm^3);
- Resistance of at least $0.000\,01 \Omega$ ($10 \mu\Omega$) in the test length between potential contacts;
- No surface cracks or defects visible to the unaided normal eye, and substantially free from surface oxide, dirt, and grease;
- No joints or splices; and
- In case of wire rod; sample may be drawn from outer side of coil.

C-5 PROCEDURE

- Clean the specimen surface for good electrical contact and make resistance measurement to an accuracy of ± 0.15 percent using micro ohm meter. To eliminate errors due to contact potential, take two readings, one direct and one with the current reversed, in direct succession;
- Maintain the consistent room temperature (recommended $22^\circ\text{C} \pm 2^\circ\text{C}$) and test specimen must be kept in room for least for 45 min before taking resistance measurement to maintain the temperature uniformity in a test specimen;
- Apply the temperature correction, if measurement deviate from the temperature 20°C by the given formula

$$R_T = \frac{R_t}{1 + \alpha_T (t - T)}$$

where

R_T = resistance at reference temperature T ;

R_t = resistance as measured at temperature t ;

α_T = known or given temperature coefficient of resistance of the specimen being measured at reference temperature T ;

t = temperature at which measurement is made; and

T = reference temperature.

For purpose of calculation : α_T (aluminium): 0.004; and

- For measurement of resistivity, use formula

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$$p_v = \left(\frac{A}{L}\right) R$$

where

p_v = volume electrical resistivity, in $\Omega \cdot \text{mm}^2/\text{m}$;

A = cross-sectional area, in mm^2 ;

L = gauge length, in mm; and

R = measured resistance, in Ω .

(MTD 07)

Publication, BIS, New Delhi